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The ability of image post-production to replace analog cameras.

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Prohlášení

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Maksim Tanuilov

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Abstrakt

Popularita touhy dosáhnout „filmového vzhledu“ už několik let stoupá. Cílem práce je vyvinout pracovní postup, který pomůže napodobit tento vzhled pomocí RAW souborů a digitálního fotoaparátu a pluginu Dehancer. Práce popisuje vlastnosti filmových materiálů a digitálních souborů RAW, které hrají klíčovou roli v procesu emulace. Aby se ověřila hypotéza, že by postprodukční proces mohl napodobit vzhled filmu za podmínky použití specifického softwaru a pracovního postupu, byl proveden experiment. Série digitálních a filmových fotografií byla vyfocena v podmínkách studiového osvětlení. Pro získání objektivních výsledků byly pro obrázky použity dva stejné objektivy a stejné schéma osvětlení. Výsledky ukázaly, že vyvinutý pracovní postup dokázal napodobit vzhled filmu, měl však určité nevýhody způsobené omezením možností softwaru. Mezi hlavní problémy patří práce s barvami a stíny. Například, během úprav v sekci Colour grading střední tóny ovlivňovaly i odstíny tmavých objektů. Proto tmavé objekty na digitální fotografii barevně neodpovídaly tmavým objektům na analogové fotografii. To naznačuje, že filmový vzhled lze napodobit. Na druhou stranu nelze vytvořit univerzální přednastavení tak, aby vyhovovalo všem světelným podmínkám a kompozicím.

Klíčová slova

Film, RAW, emulace, film look, analog, software, photo, film emulation, postproduction, post-production, photo camera, lenses, vintage lenses, vintage look, vintage image, 35mm, kinofilm, 135 film, kodak portra 400, Fujifilm, fujichrome provia 100f, provia, portra, ISO, silver halide crystals, sensitivity, film sensitivity, photo materials, studio, image representation, color grading, hsl, grain, grain emulation, halation, bloom, film bloom, Adobe Lightroom CC.

Abstract

The popularity of the desire to achieve a “film look” has been rising for a couple of years. The goal of the thesis is to develop a workflow that helps to emulate this look by using RAW files, a digital camera and Dehancer plugin. The thesis describes the characteristics of film stocks and RAW digital files that play a key role in the emulation process. To test the hypothesis that the post-production process could emulate the look of film stocks upon the condition of using specific software and workflow, an experiment has been conducted. Series of digital and film photographs have been shot in the studio lighting conditions. To receive objective results, two same lenses and the same lighting scheme have been used for pictures. The results showed that the developed workflow can emulate the film look, however, it did have some disadvantages caused by software limitations. The main problems include working with colors and shadows. For example, during editing in the Color grading section, the midtones also affected the shadows of dark objects. Therefore, the dark objects in the digital photo did not match the dark objects in the analog photo. This suggests that film look can be emulated. On the other hand, a universal preset cannot be developed to suit all the lighting conditions and compositions.

Keywords

Film, RAW, emulate, film look, analog, software, photo, film emulation, postproduction, post-production, photo camera, lenses, vintage lenses, vintage look, vintage image, 35mm, kinofilm, 135 film, kodak portra 400, Fujifilm, fujichrome provia 100f, provia, portra, ISO, silver halide crystals, sensitivity, film sensitivity, photo materials, studio, image representation, color grading, hsl, grain, grain emulation, halation, bloom, film bloom, Adobe Lightroom CC.

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1. Introduction

Analog photography has been experiencing a new wave of popularity since the second half of the 2010s. Kodak, which filed for bankruptcy in 2011, regained its position back on the market during the new renaissance of analog photography and announced the resumption of production of the discontinued film. However, the market for analog products can change again shortly. In 2021 Kodak and Fujifilm announced a future reduction of the range and an increase in the price of products in the foreseeable future. This decision relates to the production issues and problematic delivery of raw materials for the film. The threat of stopping further production of photographic film is very real due to logistical problems and its negative impact on nature. Experts predict that the film will take a more niche position for the future generation (Nico's Photography Show, 2022), and manufacturers will reduce their range exclusively to professional photo materials (like the Kodak Portra series). Furthermore, the film will be produced at the request of film studios (for example, a new season of the popular HBO show Euphoria used Kodak Ektachrome). In this regard, film stocks will become less accessible to the average user. Therefore, photographers, cameramen, and editors will be forced to look for alternative ways to reproduce the "analog image".

The purpose of this work is to find a way to achieve an analog image look by using a digital image and further post-production based on the physical characteristics of analog photographic materials. That will allow artists to achieve an analog look in their visual works using more affordable modern gear.

To achieve the goal, the following tasks must be completed:

In the theoretical part of the thesis:

- defining physical and chemical characteristics of analog medium (film stock) that influence the visual representation of an image
- defining physical characteristics of modern and most used digital sensors for the photography and their influence on the visual representation of an image

In the practical part of the thesis the experiment must be conducted:

- making a series of photos in a studio using various analog mediums (color positive film, color negative film) and modern CMOS sensor photo camera
- making digital photos look similar to analog photos by using knowledge of film stock characteristics and further post-production in two different software (Lightroom classic and Dehancer)
- finding out an algorithm of "analog look" imitation and making a workflow of the process
- analyzing the result of the experiment

The theoretical part begins by defining some of the basic terms used in the thesis: film stock, negative and positive processes, sensitivity and film grain, color temperature, color palette, and distortions.

The chapters in the practical part describe the experiment that was conducted in the studio. As a result, theoretical knowledge from the previous part of the thesis was applied to emulate two types of film.

The hypothesis for the experiment is defined as “post-production process could emulate the look of film stocks upon the condition of using specific software and workflow”.

The gear that will be used for the experiment is two Canon cameras (Canon EOS500 (Kiss), Canon 5d Mark III), two Canon lenses (Canon EF 24-105mm f/4 L IS II USM, Canon ultrasonic 28-80mm f/3.5-5.6 II (the year 1999)), studio lighting and two types of film stock (Kodak Portra 400 in 35mm and Fujichrome PROVIA 100F Professional [RDP III] in 35mm).

The goal of the experiment is to acquire RAW digital files, and negative and positive film images to use in later stages of emulating the film stocks. The next step of the experiment consists of developing the workflow for the process of emulating specific film stocks.

2. The theoretical part

In this chapter of my thesis, I am going to analyze the physical and chemical characteristics of photosensitive materials, which influence the representation of an image. I will apply this knowledge to emulate two types of film stocks using digital RAW files and the post-production process in the practical part of the thesis.

2.1 Analog media. Film stocks

The film stock is an analog medium, (IOFIS, 1980) which is used to capture an image. The base of the film is a sheet or a strip of transparent material. It is coated on one side with gelatine emulsion, which contains microscopically small light-sensitive silver halide crystals. It has different resolution, sensitivity, contrast, color palette, and methods of the development process (Keller, pp. 1-45)

2.1.1 Sensitivity of the film stock

Depending on the type of the film, one cm^3 of a film may contain 10^5 - 10^{14} micro silver halide crystals (Figure 1). They are dispersed randomly and overlap with each other. The intensity of film sensitivity is also connected to the size of crystals. The bigger crystals have more intensity than the smaller ones. Different kinds of photosensitive layers contain different sizes of crystals. (IOFIS, p. 4)

Modern-day's standard of film speed rating is called ISO. It measures sensitivity to the light. Various film stock speed scales determine the sensitivity of film stock by controlling the amount and the size of silver halide crystals on emulsion defined by algorithmic and logarithmic scales (Allen, pp. 372-374) (IOFIS, p. 4).

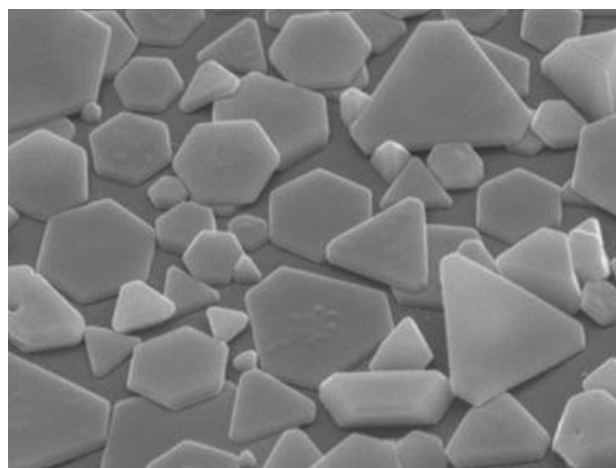


Figure 1. Silver halide crystals

2.1.2 Positive and negative film process

Film stocks are divided into slide (positive) film and negative film.

Negative film is a type of film stock that allows getting a color (or black and white) inverted

image after the developing process on a strip of transparent material. Negative film stocks can be color or black and white. The diameter of silver halide crystals in low sensitive negative stocks is approximately 0,7 micrometers (IOFIS, p. 4)

As we can see on the Ilford graphs (Figure 2), negative stocks have a wide dynamic range. The high dynamic range of negative film allows overexposing it (up to 8 stops) and still getting a usable image. However, underexposing the negative stock (over -3 stops) can cause the loss of information in shadows. Moreover, color negative stocks have small color contrast. These characteristics make an image flat looking (The Slanted Lens, 2018).

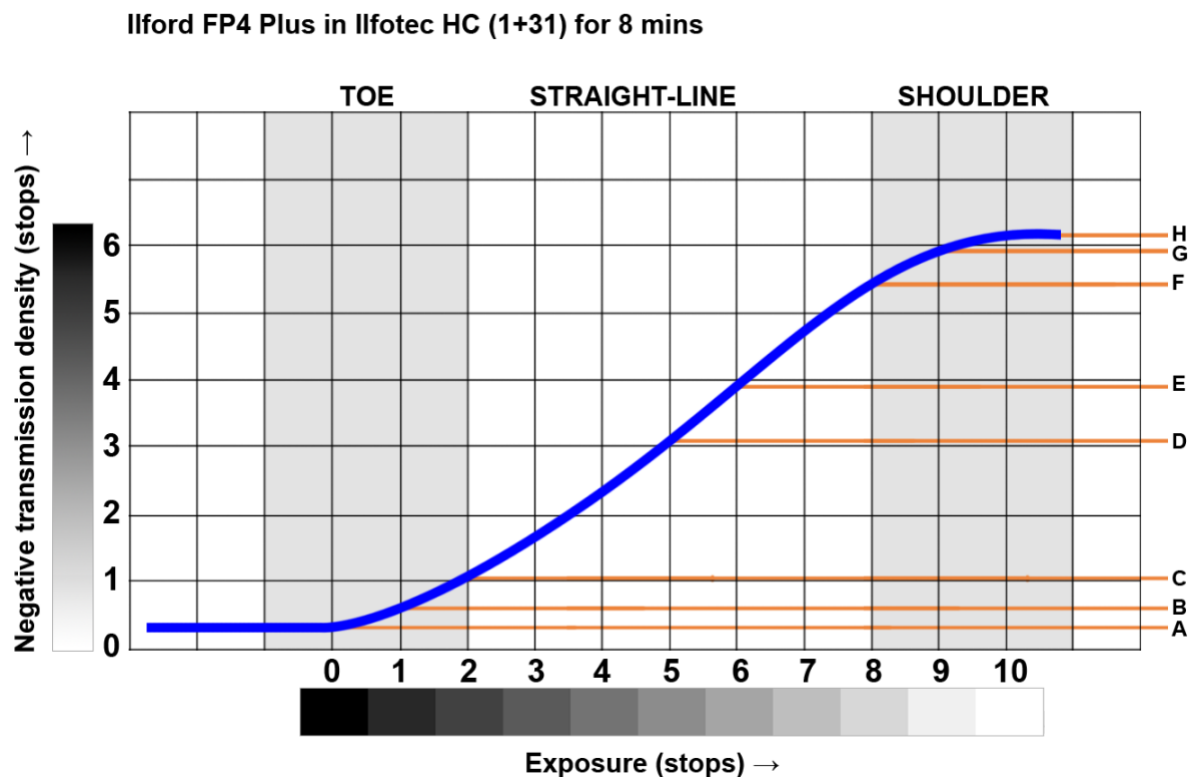


Figure 2 The correlation between exposure and density of the negative film

Slide film produces a positive image on a transparent film base after the developing process. It allows projecting an image on a screen by using a slide projector. An image made with a slide film can be black and white or color (Marquardt, pp. 84-85). The diameter of silver halide crystals that are used in low-sensitive slide film is approximately 0,63 micrometers (IOFIS, p. 4). That is the reason why reversal film produces a sharper image than negative film. Moreover, slide film produces more saturated colors in comparison with negative film (The Slanted Lens, 2018). As we can see on the graph (Figure 3), it has a low dynamic range. It can cause problems during the exposure of high contrasted compositions or incorrect exposure settings. The setting below -3 or over +3 exposure stops can lead to loss of information.

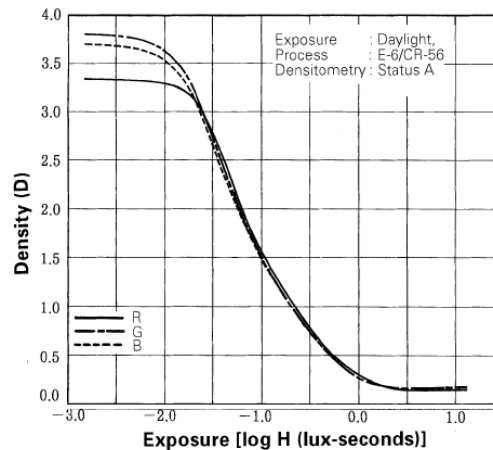


Figure 3 The correlation between exposure and density of the slide film

The developing process is an important step that can change the whole look of an image. Different films are developed using different methods. For the color negative it is the C-41 process. For slide film - E6 process. These steps use the reaction of the developer with silver to create the dye. The emulsion used on a slide film results in a color-positive image, while the combination of the emulsion and C-41 process creates a color-negative image (Marquardt, pp. 82-86).

2.1.3 Color temperature and Color palette

Color film stocks have a calibrated color balance. It is used to accurately capture the white on a picture. White color is defined by the color temperature that is set for particular film stock. For the daylight stocks, manufacturers use 5500-5600K color temperature calibration, which has a yellow tone. It was designed to be used specifically in natural light settings. For the scenes with artificial light, film stocks have a 3200K white balance calibration. It adds blue tones (Allen, p. 396).

Every film stock has its own color palette (Wilhelm, p. 15). It depends on the technology used during the production of film stock. For example, even a professional series of film stocks such as Kodak Portra has soft color tones because of the saturating issue of converting a negative image to a positive one. To find out a color palette of a particular film stock, it is recommended to refer to manufacturers' descriptions or to use a ColorChecker (Figure 4).

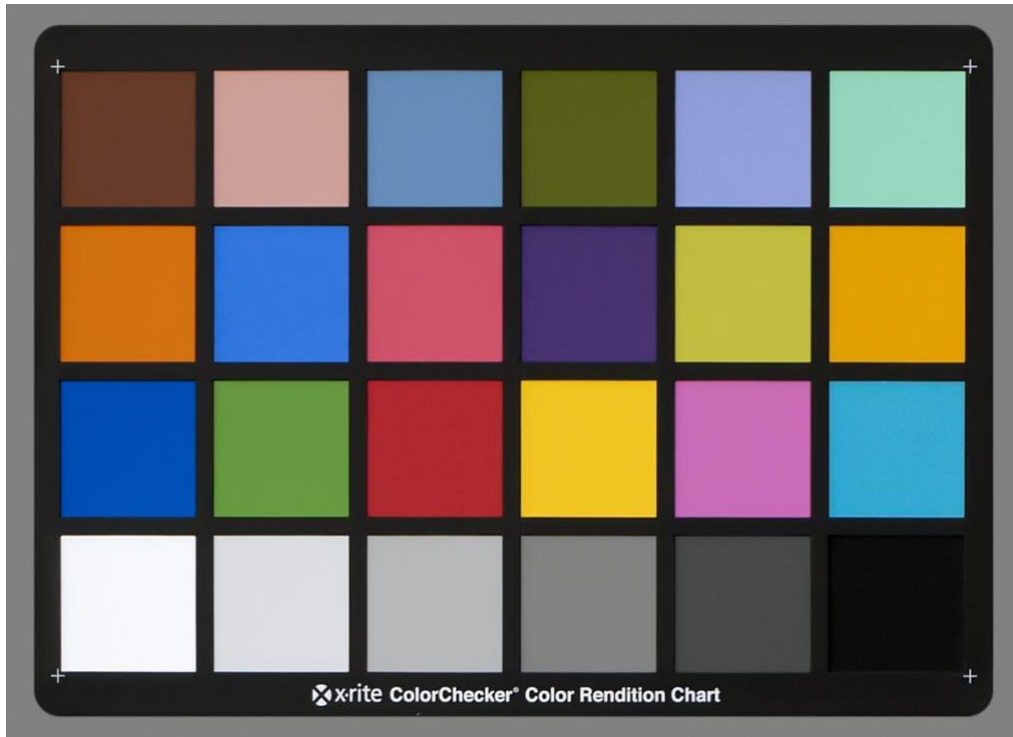


Figure 4 ColorChecker

ColorChecker is used to create a consistent look of colors under different lighting conditions.

2.1.4 Distortions, and aesthetic of the photography film

Working with the film is a delicate process. Professional film photographers always check their cameras for light leaks before shooting. Light leaks (Figure 5) can expose parts of a film before a photographer takes a picture. However, in some cases, these defects are used for aesthetic purposes to make a “nostalgic glow” in point and shoot photography (shooting process on easy-to-use cameras or disposable cameras) (Marquardt, p. 231).



Figure 5 Light leak example

Grain is formed by silver halide crystals in the process of film exposure and development (Marquardt, p. 10). As I mentioned in a chapter on “sensitivity”, crystals differ by size. Higher sensitivity films produce a less detailed but more textured, grainy image. This is the reason why professional photographers prefer to use low ISO or higher resolution films to get a sharp image. Nonetheless, film grain is often used as an artistic tool by photographers to get a less realistic but more experimental look (Figure 6).



Figure 6 Example of using the film grain as an artistic instrument by Aows

Halation (Figure 7) is a blurred glow around the edges of the highlighted area on a color film shot, which is caused by a reflection of the light through the red emulsion layer from the back layer of the film. Halation is common for the captured sources of light such as lamps or bounced sunlight. Moreover, halation influences mid-tones, which include skin tones. It slightly adds red tones to the skin of the face in scenes with the bright light. To avoid the intensity of this glow, manufacturers add an anti-halation layer. Glowing edges are common for motion picture film stocks. In photography film stocks, the halation is almost invisible due to the thick anti-halation layer (Dehancer, 2021). Nevertheless, there are film stocks for the photographers (Cinestill 50T, Cinestill 800T) that allow them to get a halation effect for artistic purposes.



Figure 7 Halation may be seen of edges of windows or skin

2.2 Digital sensors as an alternative to analog photography

A digital sensor plays a crucial role in the thesis. It allows the experiment to be conducted. However, the development of the sensor has been reliant on technological progress. From the 2000s the sensors became a part of average user life and have developed ever since (Allen, p. 10).

2.2.1 The representation of a digital image

A digital image is represented by a grid of non-overlapping pixels. They have specific coordinates in terms of rows and columns and can be represented as a scale of integers (Allen, p. 12).

2.2.2 CMOS sensors

CMOS sensors are silicon-based technology that amplifies the charge and converts the analog signal to the digital one. Each pixel is read off individually, which creates more data. Higher amounts of data lead to more opportunities during editing. Another advantage of the CMOS sensor is its power efficiency, which is important, especially in a digital camera, for shutter lag and refreshes time (makes it possible to shoot in burst mode) (Allen, 2011, p. 11).

Photo camera manufacturers are experimenting with more flexible and affordable CMOS technology. They aim to make an affordable photosensitive material that captures high-resolution images with a good amount of data to better further post-production editing.

2.2.3 Digital Image capturing

In the next paragraph, I will describe the principle of capturing digital images. This will explain the limits of the digital file to change. Furthermore, it reveals the defects that occur during the modification. Particularly crucial for this thesis is a paragraph about the color system and bit depth.

2.2.3.1 Digital noise

Every electronic device receives some amount of noise while getting a signal. Digital noise has a different appearance from the film grain. The latter has a variety of shapes and sizes. The grain is more noticeable in midtones. Digital noise, on the other hand, is more visible in shadows. The noise is unavoidable, however, if the signal-to-noise ratio (SNR) (Figure 8) is high, the distortion of an image becomes non-existent. One of the possibilities to prevent the appearance of noise in the post-production process is to create the right exposure while shooting. (Cambridge in Colour, 2022)

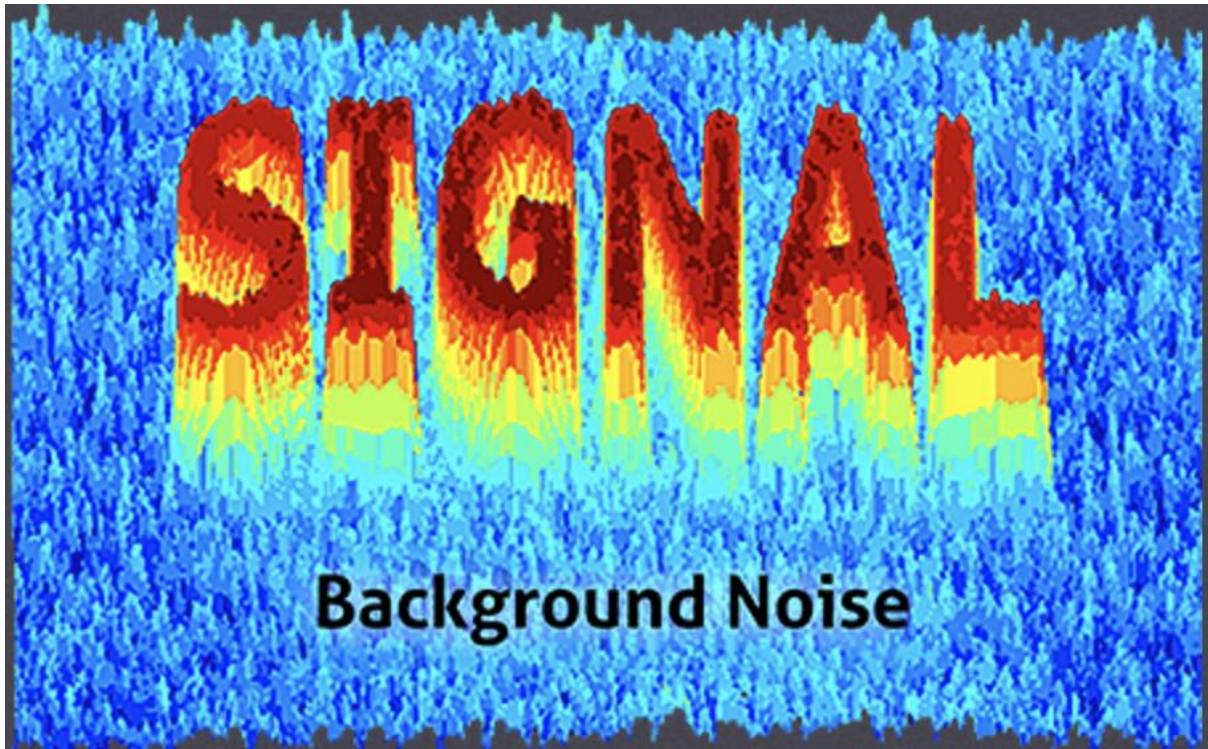


Figure 8 Visual representation of signal to noise ratio

2.2.3.2 Color system

To produce a digital color image modern cameras are using sensors with overlaid colored filters. The sensor is capturing an image by using a Bayer pattern (Figure 9) and then placing red, green, and blue pixels to correspond with the original scene.

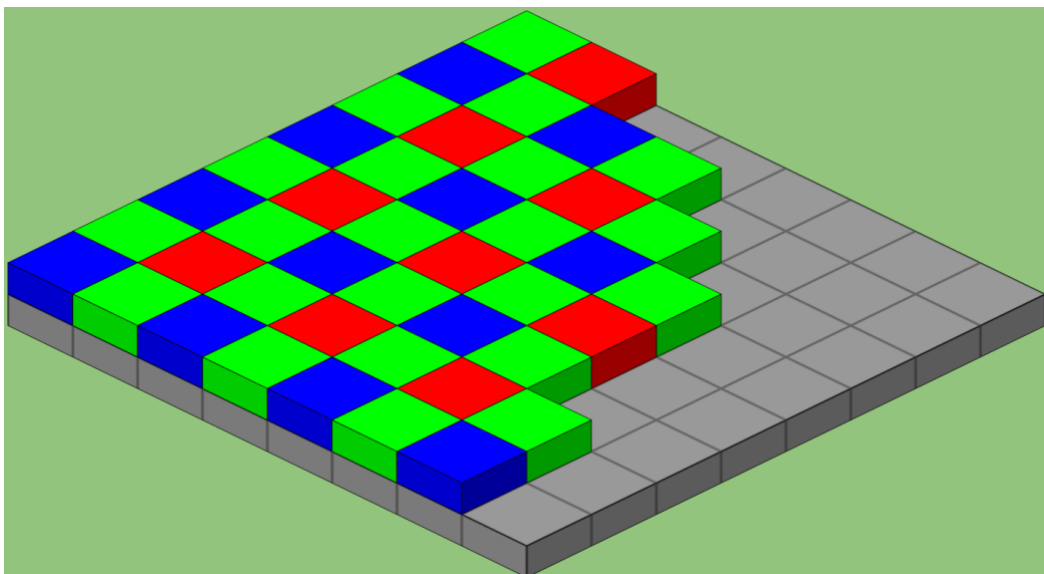


Figure 9 Bayer pattern

There are twice the amount of green pixels in comparison to red and blue ones in the array. This is caused by the effort to recreate the human visual system. Its peak sensitivity is to the

green light. This means that the human eye can register more errors in the green channel (Allen, p. 11)

2.2.3.3 Bit depth

Bit depth shows how many colors are accessible in an image's color palette in terms of the number of "bits," which are used to specify each color.

High numbers of the bit depth include more colors and have a larger digital file size (Allen, 2011 p. 14). Different file formats are able to use different maximum bit depths. TIFF can display 16 bit, while JPG only 8. One can use a higher bit depth, however, not a lot of formats are able to display it. "On the other hand, images with more than 24 bpp are still quite useful since they hold up better under post-processing"(Cambridge in Colour, 2022). The more colors are available, the better transitions from shade to shade are available. This is one of the biggest differences between analog and digital images (Figure 10) (Allen, pp. 286-287). This information is used later in the post-production phase.

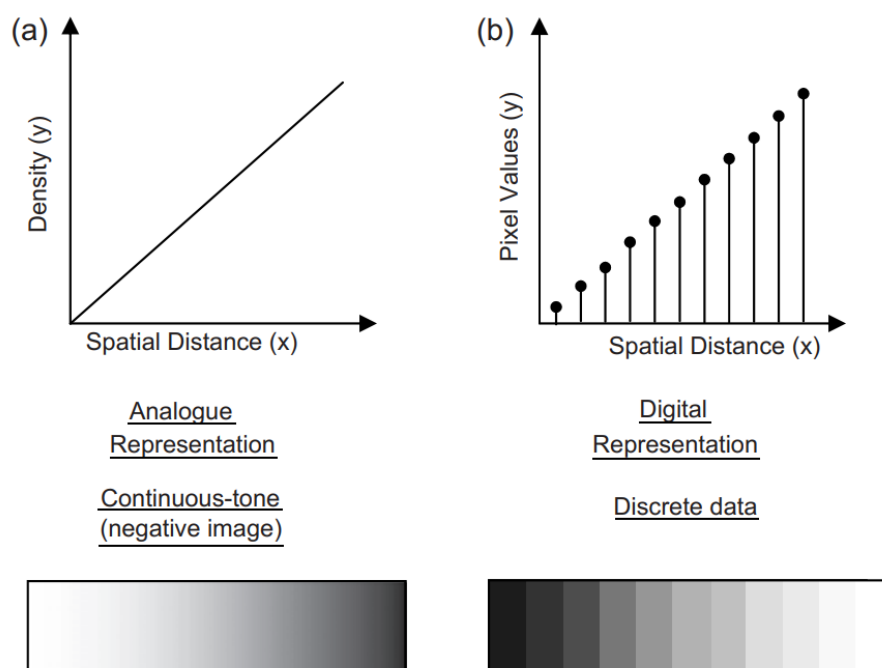


Figure 10 Difference in representation of transition from the darkest to the brightest areas

To sum up this part of the thesis, I would like to underline that this knowledge is a crucial part of the post-production process. To emulate the film look, characteristics of the film stock are essential. Among the most significant are film grain, halation, and density. The knowledge about the flexibility of digital files plays a key role in achieving satisfying results.

For example, the difference between the transitions from shade to shade in analog or digital files. The significance of digital noise and the necessity of its reduction.

In the next chapters, an understanding of the key characteristics will be applied during the practical experiment.

3. The practical part

To test the hypothesis, an experiment has been conducted. This method of research was selected due to its practical approach; it provided an opportunity to fully control the quality of source images. In this chapter, the thesis describes the experiment's process, medium, and equipment. To indicate whether a film look could be reproduced on a digital image, a set of pictures was taken on both film and digital cameras. Applying the knowledge from the theoretical part, a universal workflow for emulating a film stock has been created.

To keep the results of the experiment as objective as possible following adjustments have been made: the same studio lighting and position of the camera have been used for both film and digital photographs. Moreover, the same lens was specifically used on both cameras to test the photosensitive materials.

3.1 Equipment and medium of the experiment

To make the workflow universal, affordable, and accessible gear has been used. Two Canon cameras with the same EF bayonet were selected for the experiment (an analog one and a digital one). An analog camera was Canon EOS500 (Kiss), and the digital one was Canon 5d Mark III (with CMOS sensor). The latter, according to my research, is considered to be one of the most accessible full-frame DSLR cameras nowadays (Roy Riley, 2021).

The first lens used for research was a modern Canon EF 24-105mm f/4 L IS II USM, the other one was a vintage Canon ultrasonic 28-80mm f/3.5-5.6 II (year 1999). The choice for the two lenses was justified by their accessibility on the market. Two professional film stock series were used to perform the experiment: Kodak Portra 400 in 35mm and Fujichrome PROVIA 100F Professional [RDP III] in 35mm (Eastman Kodak, 2022), (Fujifilm, 2022)

The experiment took place in a photo studio. To achieve satisfactory results two DIGITAL MASTER PRO DM-4 with a softbox and a large octabox were utilized to light different compositions and a ColorChecker to calibrate white balance and colors.

3.2 Objects in the photos

To test the ability of the analog medium and a CMOS sensor to deal with overexposure and underexposure, various series of compositions with different light intensities were designed. The choice of the surfaces (glass, fabric, paper, rubber) was made to test which of them had more potential to be realistically replicated by using a digital image and the post-production workflow.

The first composition was the still life lit with a big octabox and light reflector. The still life included various materials: a glass bird, a metal object, a painted wood figure, fabric, paper, plastic, and others. The reason for this choice is to understand how different photo materials capture various surfaces in the picture.

The second composition was a portrait lit with an octabox and softbox. This choice of equipment was made to test the ability of photosensitive materials to represent human skin and test color characteristics in the correct exposure situation.

The third one was a portrait with a difficult light exposure situation. The correctly exposed forefront has allowed for the correct exposure of the right side of the face. However, the overexposed background was reflected on the left side of the face, which caused it to be too bright. This decision was made to test the density of the different photo materials and to test their ability to deal with difficult light situations.

3.3 The process and raw results

The main and secondary light were put to use to conduct the experiment: the first one (DIGITAL MASTER PRO DM-4 with big octabox) with 8.0 intensity and the secondary one (DIGITAL MASTER PRO DM-4 with small softbox) with 6.0 intensity. Also, the silver reflector was used to light the still life. Camera settings were f/16 (and f/20) aperture with a 125 ms shutter speed and 100 ISO, 400 ISO.

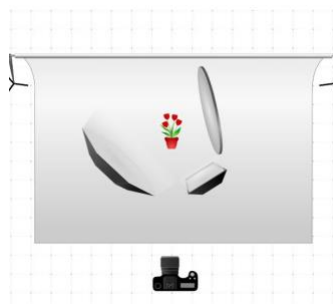


Figure 11 the still life light settings

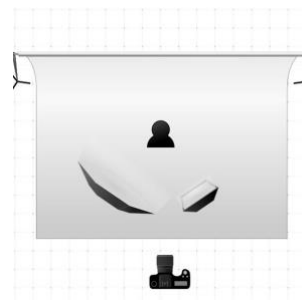


Figure 12 the portrait (normal exposure) light settings

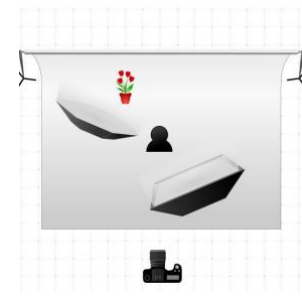


Figure 13 portrait (extreme exposure) light settings

The intermediate results were quite interesting. To get them an Epson v600 scanner was put to use to see how the film came out.

Film photos do need further post-production to achieve satisfactory results. Even darkroom printing needs color calibration to get the results that a photographer needs (Willem Verbeeck, 2019). The same principle can be applied to the process of digitalizing a film. Raw scans come out flat and less color-saturated no matter what scanner is used. (William Sheepskin, 2022). However, this does not mean that one should not analyze Raw scans or film. Even a raw image can offer valuable information before the emulation process.

3.3.1. Slide film (Provia 100f) intermediate results

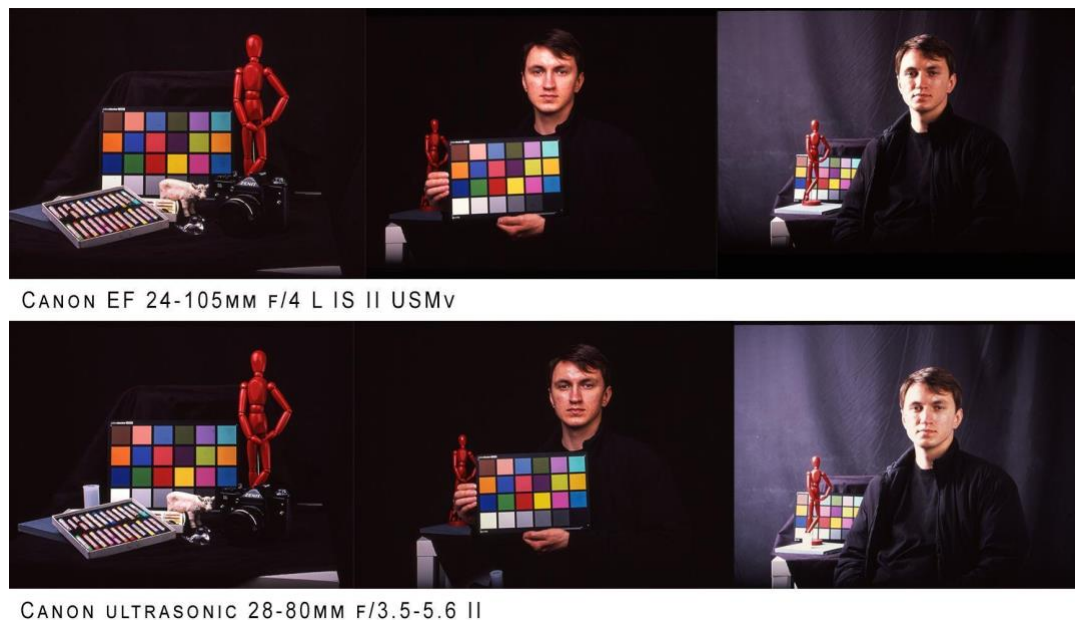


Figure 14 Raw slide film results using different lenses and lighting

Raw scans of color slide film came out extremely contrasty. Every picture has very saturated colors and rich tones. Frames on slide film have a small amount of grain and do not look textured.

In the still life, the background is black and lacks information. Also, there is a huge contrast between shadows and highlights on main objects (especially with modern Canon EF 24-105mm lens).

The portrait with correct exposure came out very saturated, skin tones are slightly reddish in comparison with the reality. However, the jacket on the portrait came out very dark (even though it was lit with secondary light).

The portrait with a difficult exposure situation came out unsatisfactory. Slide film can not handle the lighting conditions, there is a huge contrast between highlighted areas and shadows. Even post-production would not help fix this problem.

In conclusion, it should be noted that the experiment proved statements and facts about slide film discussed in the theoretical part of the thesis. Images came out very saturated and contrasted but with poor dynamic range.

Frames taken with a modern lens have much more contrasting colors and shadows. A vintage lens gave the photos a more flattish, old photos look.

3.3.2. Negative film (Portra 400) intermediate results

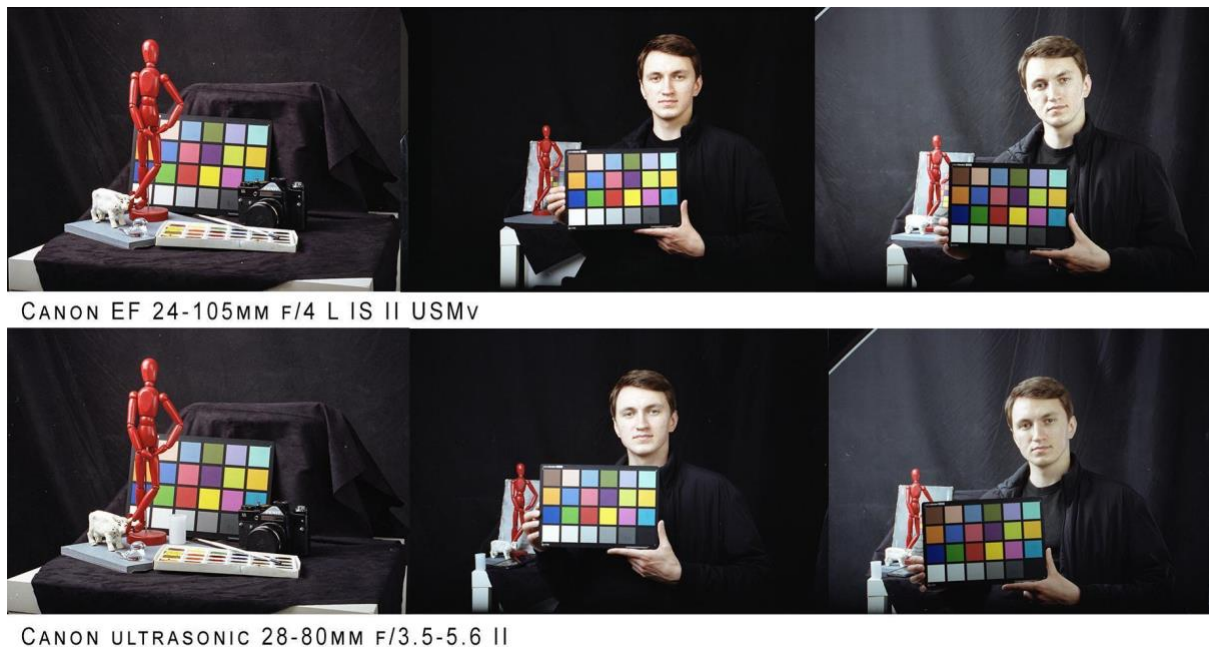


Figure 15 Raw negative film results with Kodak Portra 400 using different lenses and lighting

Raw scans of color negatives also have fascinating results. Images came out flat, with desaturated colors but with an extremely wide dynamic range. Portra 400 has a more grainy look in comparison to a slide film, however, it is not as textured as unprofessional film stocks. In the still life image, the whole frame is equally covered with ambient light (even the background). There are impressive transitions between highlights and shadows, and a huge amount of midtones are visible on the fabric, but the colors are faded. ColorChecker looks washed out, especially in bright tones such as “Light skin”, “Neutral 8” and “Neutral 6.5”.

The portrait with correct exposure came out bright and flat. Unlike slide film, color negatives are brighter; the jacket is exposed correctly and does not blend in with the background. Skin tones are desaturated, look unnatural and require further post-production to get correct results.

The results of the portrait with a difficult exposure situation are very impressive. Due to the wider dynamic range, color negatives easily handle +3 stop overexposure on the background. The image came out brighter than normal exposure but it can be fixed in post-production without any problem.

To summarize, a color negative film is a very flexible medium. It has impressive dynamic range, sharpness, character, and the ability to be modified during post-production. The photos, in which a modern lens was utilized, appeared to have slightly more contrasting colors and shadows, but the lens did not affect the image as much as in the case of the slide film.

3.3.3. RAW digital intermediate results

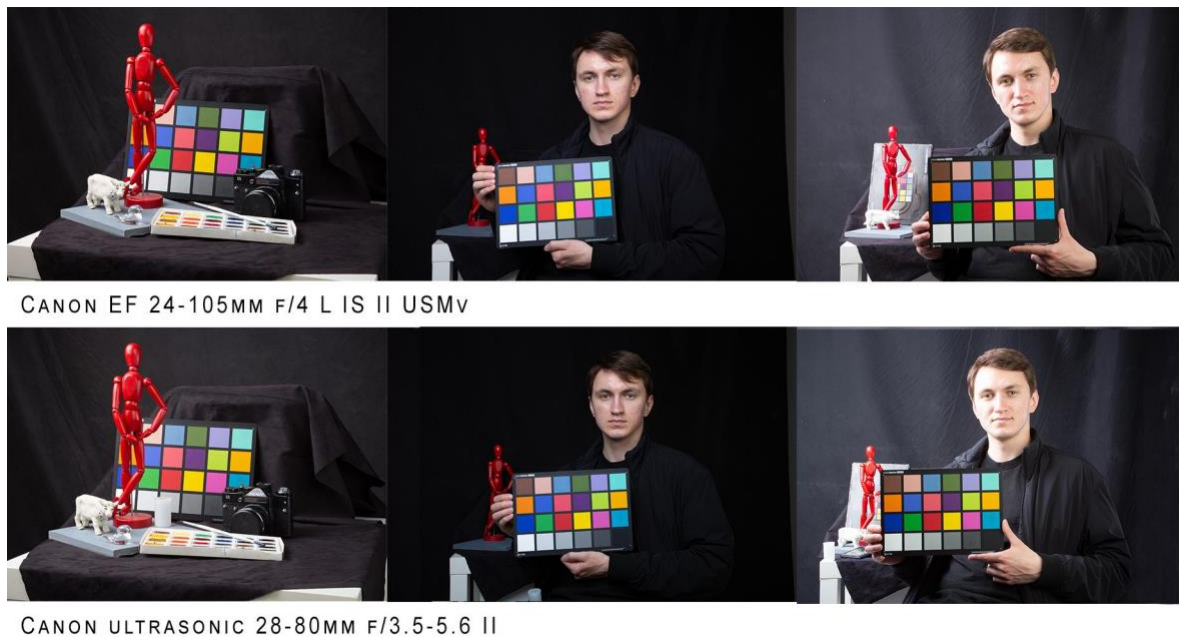


Figure 16 Raw digital intermediate results using different lenses and lighting

The look of a RAW digital image could be described as a middle ground between color negative film and color positive film. It has rich color tones like a slide film and a wide dynamic range of color negative film. It gives the artist an opportunity to influence the final results with greater digital files flexibility.

The other valuable observation is the ability of CMOS sensors to capture an image is influenced by the lens. Digital images shot on a vintage lens have a drastically different look in comparison with modern ones. The vintage lens makes images' shadows and colors appear flatter. Those characteristics may help an artist to apply a "color negative look" to digital photos even before post-production. However, vintage lenses are less flexible during difficult exposure situations (as seen in the previous images), but the amount of RAW digital data gives an opportunity to overcome this disadvantage using the software.

The development of CMOS sensors is innovated year by year. To imagine that a digital camera could handle a portrait with a difficult exposure situation seemed impossible. The experiment showed that results could have a satisfactory potential. The image looks overexposed, but the file still has the data and can be fixed in post-production.

This test shows that the process of emulating the film starts at the shooting stage. Correct exposure is more important than the post-production process. The lighting of the area must correspond to the final conception of the emulated image. For example, in the case of positive film stock, difficult exposure situations should be avoided (more than 3 stops of light

differences in one composition). In the case of color negative film, one should avoid high contrasts and underexposure. Getting correct exposure in RAW digital images is a key to perfect film emulation.

3.4 Post-production workflow

There is a considerable amount of digital software for editing photos. One of the most popular is Adobe Lightroom Classic. Using Lightroom gives the ability to edit light, shadows, and color channels. Nonetheless, the program has its limitations. That is the reason why additional software needs to be utilized to achieve a film look. Using the Dehancer plugin for lightroom gives more instruments for adding small details which have an important role in emulating the film. The plugin is easier to control and has a more user-friendly interface than other SW. Furthermore, Adobe Photoshop should be used to polish the final results. However, the experiment showed that limiting oneself to a specific software is not necessary. The workflow is applicable to different software that can change Tone, Presence, HSL, Color grading, and Tone curve.

The process starts with Adobe Lightroom. First of all, a primary preset which includes all basic characteristics of the particular film stock needs to be made. It is important to analyze the histogram of film photos before emulating. Analyzing the histogram of a reference film photo with a similar environment and light situation before the emulating process is a crucial step. To show the process of creating an emulation of a Kodak Portra 400 scanned on Epson expression 12000xl in 2048×3088 resolution, the thesis describes the needed steps.

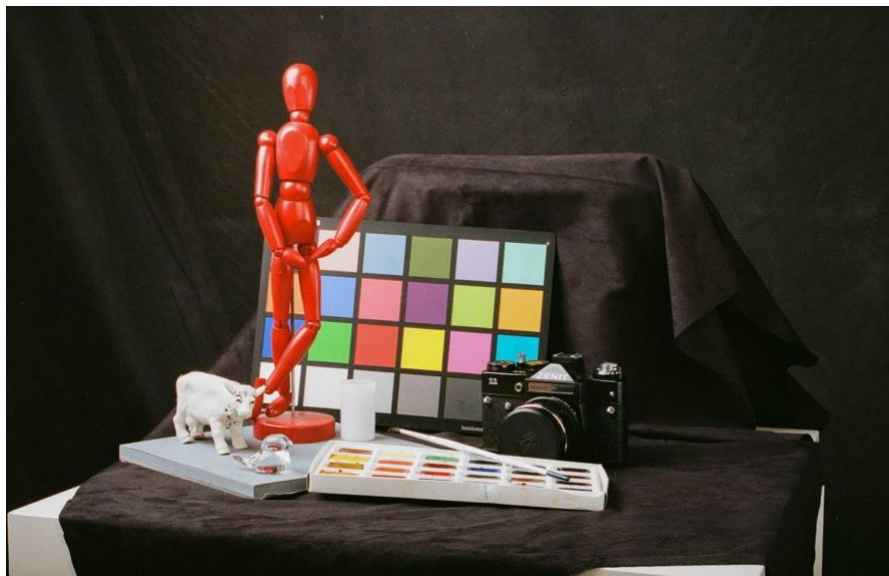


Figure 17 Kodak Portra 400 shot used as the reference

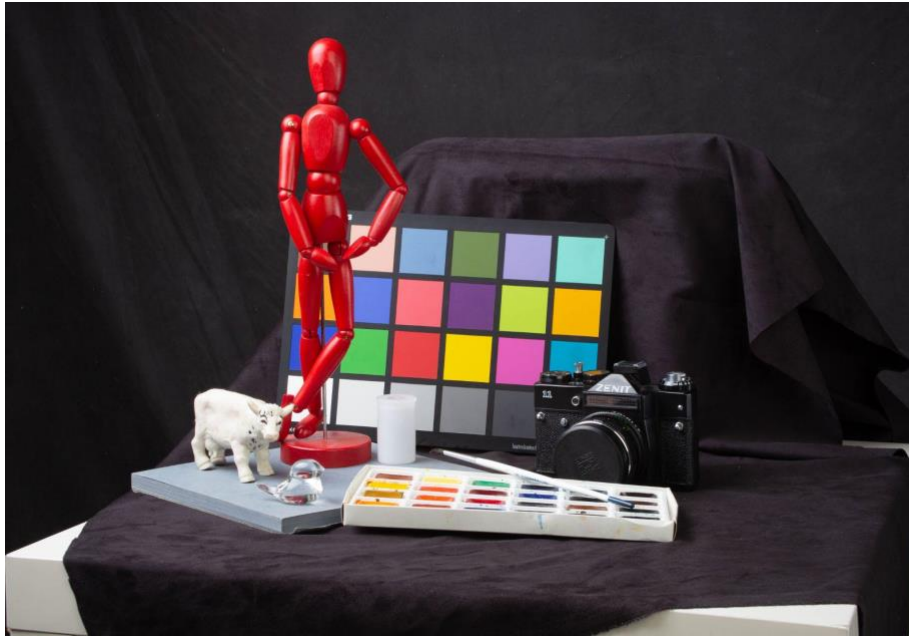


Figure 18 Raw digital not edited

A picture of the still life in the studio was chosen as an example of a film emulation workflow.

The beginning of the process



Figure 19 Kodak Portra 400 histogram



Figure 20 Digital histogram

Comparing Kodak Portra 400's histogram (Figure 19) to the digital one (Figure 20) shows film limitations in the darkest and brightest zones (it is the reason why negative films look flatter than other photosensitive materials). This knowledge becomes a base for the first step of the process of emulating: muting highlights and lifting blacks on the tone curve. Calibrating Highlights, Shadows, Blacks, and Whites are the next steps. For the picture in the experiment the following settings have been applied (Figure 21). highlights were lowered (-10), shadows were increased (+41), whites were decreased (-15), and blacks were added (-45) (to compensate for excessive flatness). Settings of "Presence" for now should be left by default. It is obvious that those characteristics depend on exposure and composition and in other cases

numbers can change. The main goal of emulating Portra 400 or other negative film is getting a flat image with smooth transitions in shadows and midtones.

Settings	Value
Highlights	-10
Shadows	+41
Whites	-15
Blacks	-45

Figure 21 Table of settings

3.4.1 Working with the color palette

The next step includes working with the colors and temperature of the picture. The Kodak Portra series is famous for its natural look. The color temperature of Portra is around 5500K - 5600K and it has an almost invisible green tint. However, as a color negative film stock, Portra has desaturated colors. It means that the process of emulating must include color grading and HSL settings. To begin the process, these terms should be explained. Color grading or color correction is the process of improving an image using midtones, shadows, and highlights. It is usually done for creative expression (HAINE, 2020). HSL calibration, however, affects the hues of particular colors. It does not change all the pixels containing a particular hue, but hues that appear a certain color to the eye. (Allen, 2011 p. 413). For the process, hues should be changed first, color grading comes next.

The first step is correcting the color temperature and tint. The white balance must be set to 5650K and a -1 green tint must be added. Moreover, after analyzing the ColorChart and density scale of Portra 400 HSL preferences are set to (Figure 22)

Color	Hue	Saturation	Luminance
Red	+14	-17	+6
Orange	+8	0	0
Yellow	-3	-1	+37
Green	-4	+13	+5
Blue	-20	-2	0
Aqua	0	-30	0
Purple	0	-6	0

Magenta	0	-9	0
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Figure 22 Table of Settings HSL preferences

Such color disbalance is happening because every color layer of a film has a different color density (Red - lower, Green - middle, Blue - Higher). Nonetheless, the whole color gamma of a Portra 400 is slightly desaturated. It means that Vibrance (-1) and Saturation (-7) should also be decreased.

One of the most important things in the film emulation process is Color Grading. Negative film stock emulsion has a green tone in shadows after converting it to positive. Professional series are famous for their natural tones but they still have slightly greenish tones in shadows. It means that in the color grading window (Figure 23) Shadows (Hue 76, Saturation 20, Luminance +42) should be set. Doing that adds a slight green tone to shadows and decreases excessive contrast by influencing luminance in the darkest areas. To affect the midtones of the picture, warmth must be added. The settings of Midtones used in the example picture are (Hue 24, Saturation 13, Luminance +45). This orange tone compensates for the greens added before. The highlights in color grading should remain unedited. Settings for blending is 25, and for Balance is +22.

Adjustments	Hue	Saturation	Luminance
Shadows	76	20	+42
Midtones	24	13	+45
Highlights	0	0	0

Figure 23 Table of settings for color grading

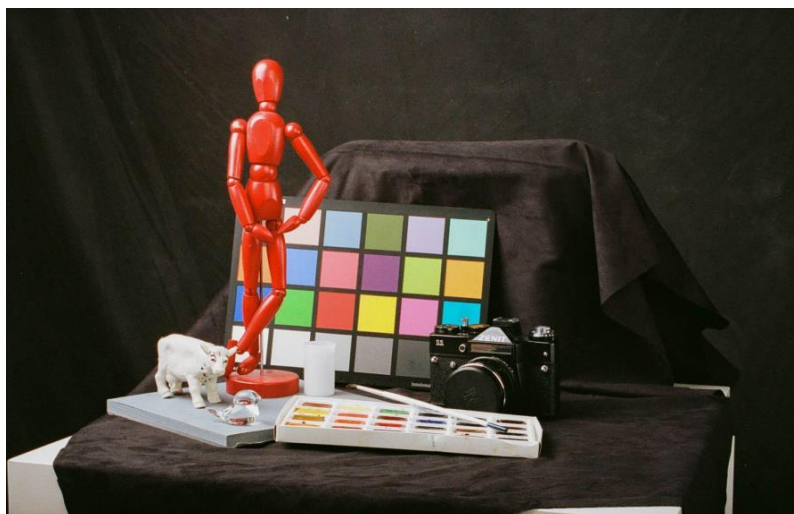


Figure 24 Kodak Portra 400 reference



Figure 25 Intermediate results of post-production

At this point, the primary lightroom preset is done. The color correcting steps are done.

3.4.2 Working with the Dehancer plugin

The next portion of the process consists of working with Dehancer. It is the most crucial process of emulation. It helps to add small but very important details that make a digital image look like an analog one. However, adding those details is a delicate process that requires skill and experience in working with analog photography. That is why keeping a reference picture is recommended while working.

There is an option to open Dehancer as a plugin straight in Adobe Lightroom Classic. The RAW file must be converted into another format. To save the maximum amount of information and quality of a picture, it must be converted to TIFF format and sRGB color space. As color scientist Karl Lang said: “sRGB is a fundamental way of communicating color in our modern digital age”. Choosing this color space ensures most of the devices will represent the color in a way an artist desired.

The higher the Bit Depth, the more colors will be available to edit. However, the size of the file would exponentially increase. (Allen, 2011 p.463-470)

Dehancer offers different profiles of film emulation. However, making a profile based on the reference brings the most accurate results.

The workflow in Dehancer starts with a Film Grain section. Just as during the oil painting, the process starts with layering and gradually adds small details to achieve a realistic look. Grain texture brings characteristic detail to the film image. Dehancer has one of the advanced grain imitations. The plugin does not overlay an image with the texture of film grain. It allows the generation of Grain in real-time and influences its Size, Amount, Resolution, Shadows, Midtones, Highlights, and Color. In the Kodak Portra 400 case, the grain is fine even though it is 400 ISO. However, there is no strict formula for adding grain. It is on the author to decide the right amount according to the desired look. According to the experiment’s reference (not cropped frame, 3 by 2 aspect ratio, shot and developed at box speed) (Figure 26) Size is set to 3.3, Amount - to 35, Resolution - to 50 (to get rid of sharpness of digital sensor), Shadows - to

30 (According to the characteristic of Portra to produce fine grain), Midtones - to 39 (to add micro contrast), Highlights - to 10.5 (In the lightest areas silver crystals are exposed without distortions. Grain is more characteristic in zones of light and shadow transitions), Color is set to 0.

Settings	Values
Size	3.3
Amount	35
Resolution	50
Shadows	30
Midtones	39
Highlights	10.5
Color	0

Figure 26 Settings for grain section in the Dehancer plugin

The next step includes working with the Print section. It is emulating printing paper. Dehancer allows emulating the printing process. It enables restoring contrasts and influences color density. Using that section is optional. During the editing of the picture used in the experiment, slight Tonal Contrast has been added and Color Density has been decreased after film grain generation. The settings consist of: Kodak Endura Glossy Paper, Tonal Contrast is set to 4, Color Density - to -2, the remainder is left by default.

The workflow continues with the addition of the most important details. As mentioned in chapter 2.5 Halation is a blurred glow around the edges of the highlighted area on a color film shot, which is caused by a reflection of the light through the red emulsion layer from the back layer of the film. Portra 400 has a powerful anti-halation layer but it does not absorb 100% of the light. In the examined picture, halation (Figure 27) can be seen in the brightest areas and edges of the objects. Nonetheless, there are more places where halation can influence the representation of an image. It can be a bright area that reflects the light. In the case of the examined picture, it is on a watercolor palette and the brightest areas on the fabric.

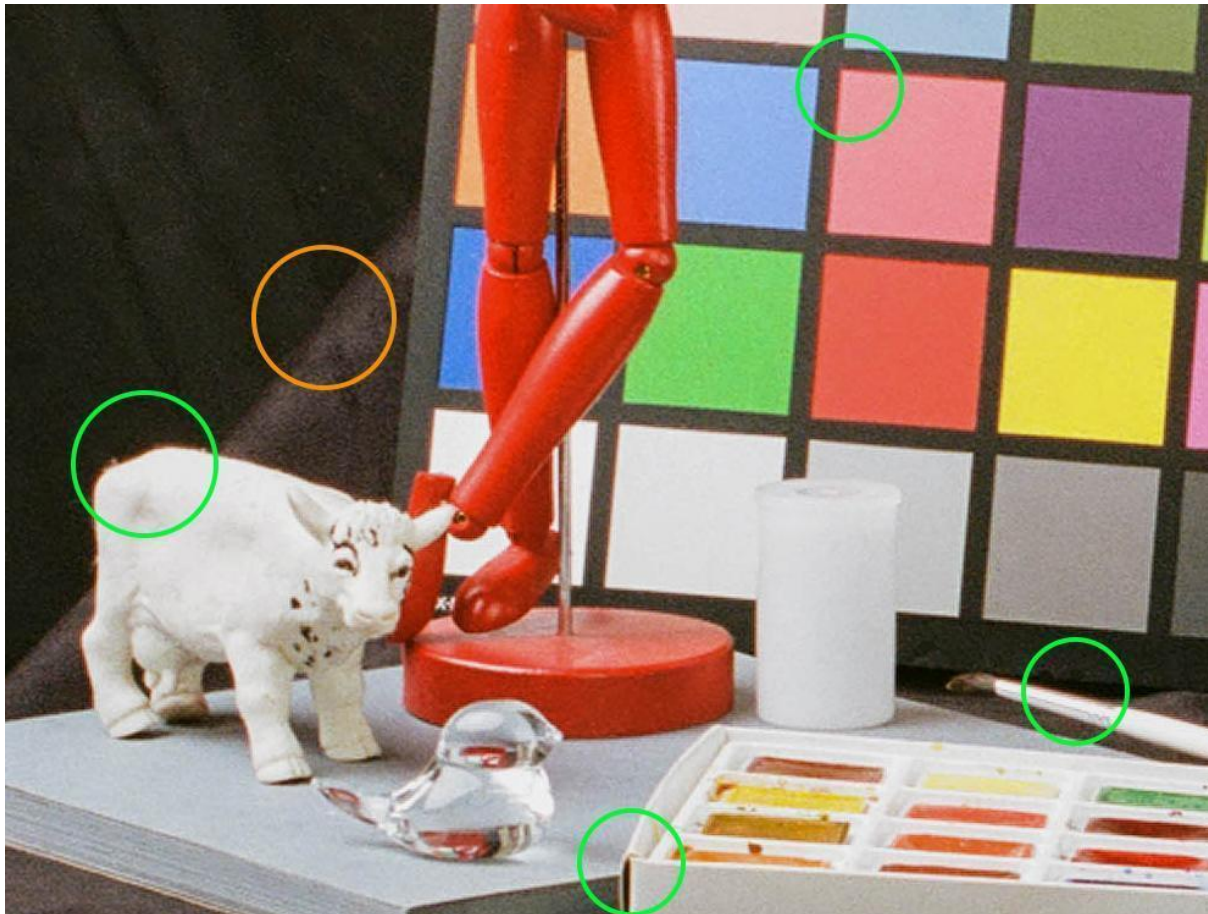


Figure 27 Examples of halation on Kodak Portra 400

Detail of a Kodak Portra 400 shot. Green circles show spots where strong halation can be observed. The orange circle shows halation on midtones in bright areas.

Before applying this effect, the intensity of halation must be identified. There are films like Cinestill that have very strong halation but most of the films have an antihalation layer. Kodak Portra 400 has very low halation in comparison with other film stocks. The setting (Figure 28) in this section is: Source Limiter is 2.2, Background Gain is 60, Smoothness is 17, Local Diffusion is 21.5, Global diffusion is 25, Amplify is 16, Hue is 54, Blue Compensation is 0, Impact is 43.5. Using these preferences achieves a low but global halation on the whole image. For the controlled spreading of halation, it is recommended to use Mask Mode. It allows seeing just the impact of any effect (Figure 33).

Settings	Value
Source Limiter	2.2
Background Gain	60
Smoothness	17
Local Diffusion	21.5

Global diffusion	25
Amplify	16
Hue	54
Blue Compensation	0
Impact	43.5

Figure 28 Settings for halation section in the Dehancer plugin



Figure 29 Before the halation was applied



Figure 30 After the halation was applied

The next important detail is Bloom. This effect allows adding light reflection from objects and adds more contrast to the highlighted area.

The primary preset creates a flat effect now. To change that, micro contrast must be restored in particular areas. This will create an illusion of a higher dynamic range. The main aim of the Bloom section is to increase the intensity of the brightest areas.

These settings were applied to the examined picture (Figure 31):

Settings	Value
Highlights	12
Source Limiter	2
Details	8
Diffusion	2
Amplify	30
Save Lights	15
Saturation	30
Impact	16

Figure 31 Settings for the Bloom section in the Dehancer plugin

Using the Mask Mode shows the result of the Bloom effect applied to the picture.



Figure 32 Before Bloom



Figure 33 Mask mode (using Bloom)

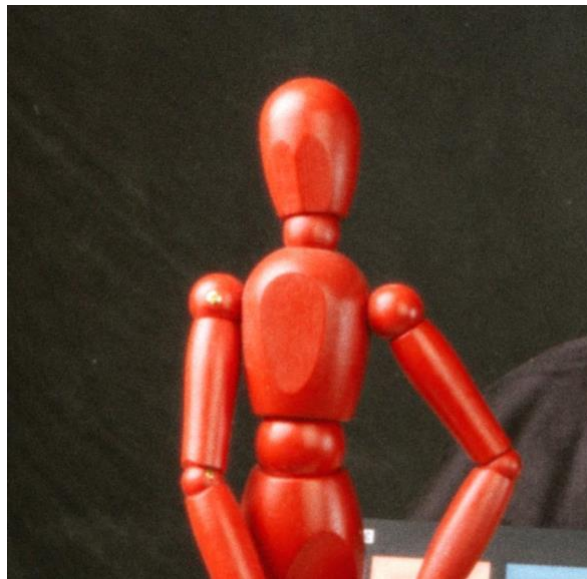


Figure 34 After the Bloom was applied

Work with Dehancer is done. Current settings can be saved as a preset for further emulations. All edits at this point must be final because the image will be converted into a new file and any change will not be available.

The next step of the workflow includes the Presence section In Adobe Lightroom Classic. For the image used as an example, Texture and Clarity were set to +16 to make the grain more visible and restore the sharpness of the image. To compensate for the increased contrast, Dehaze was set to -5.

The final steps consist of digital Noise Reduction and Final edit of the Shadows section. To eliminate digital noise, the Detail section in Lightroom must be opened, In Noise Reduction parameter Color must be set to 48. Other preferences should remain unedited. The examined image came out to be highly contrasted. This can be fixed by increasing the shadows (+8) and decreasing highlights (-20).

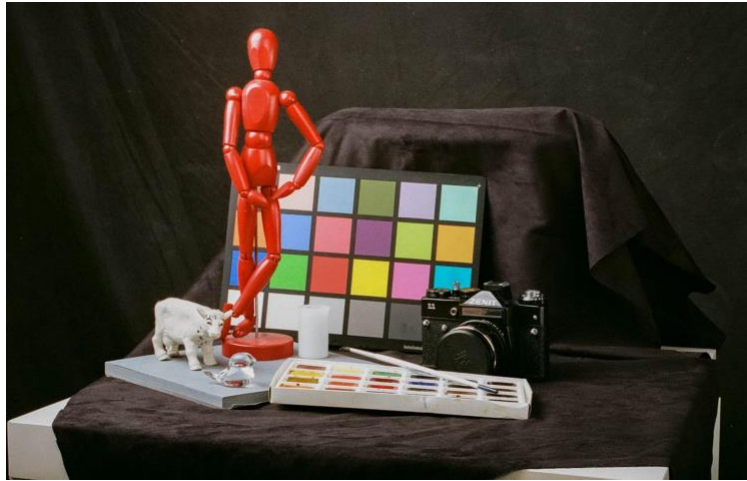


Figure 35 Kodak Portra 400 photo

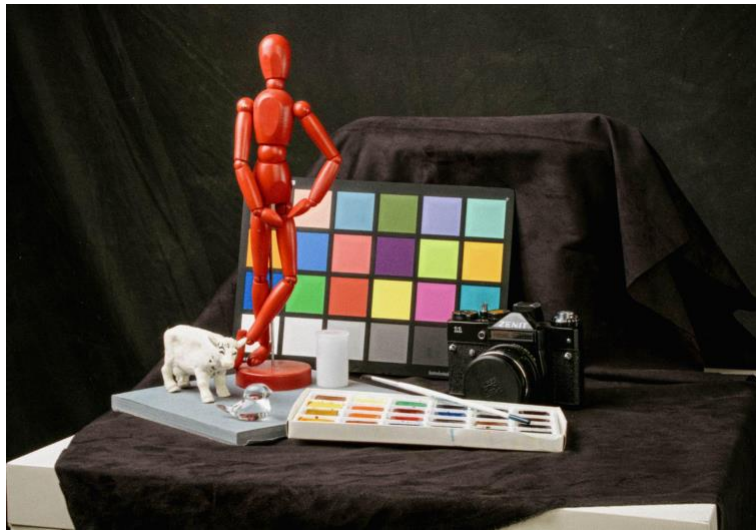


Figure 36 Digital photo after the post-production process



Figure 37 Kodak Portra 400 photo (skin tone sample)

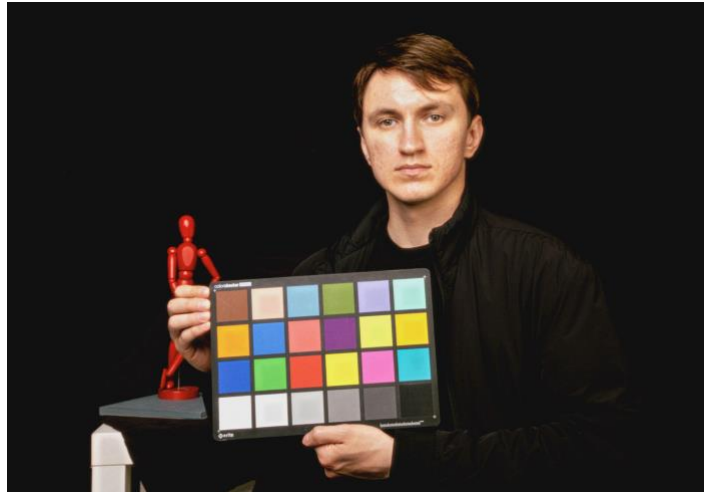


Figure 38 Digital Photo after post-production (skin tone sample)

3.5 Control of the correctness of the workflow

In this part of the thesis, the workflow will be applied to emulate Fujichrome Provia 100f (Figure 39). This part will not contain details and numbers. The main goal of this chapter is to prove that methodology can realistically emulate the film look.

One should remember two fundamental rules – starting the process with finding a reference and thinking about the composition and final image look before the shooting. Fujichrome Provia 100f is a slide film. This means editing will be based on the physical and chemical characteristics of this particular film stock. Slide film has high contrasts and low dynamic range. Those characteristics as well as color representation are similar to CMOS sensors (Figure 40).

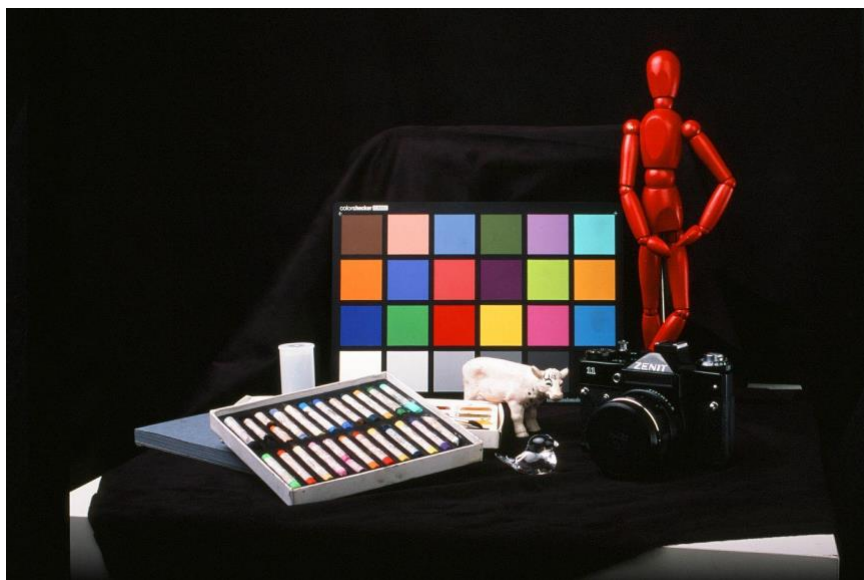


Figure 39 Fujichrome 100F photo



Figure 40 Digital RAW photo

The post-production process starts with analyzing a histogram. RAW digital shots have more information than any film shot. This information should be used for editing according to the desired final look. That is why highlights should be slightly muted on the Tone Curve and blacks should be lifted. To achieve a slide film look, a strong contrast should be created. It can be done by decreasing shadows and increasing Highlights. Blacks and Whites should be used to compensate for micro contrasts.

The next step involves changing colors. The color palette of a slide film is closer to reality than negatives but still has its own character of color representation. It is a daylight film therefore white balance is around 5500K and has no tint. According to the ColorChecker, colors look very natural and close to the RAW digital. Nonetheless, it still needs calibration. Reds have a slight orange hue, Purple has a slight violet hue, and Aqua has a deeper (dark) tone. The other colors look the same as on a digital photo. The final step in color calibration is color grading. Fujichrome Provia has almost no specific tones in shadows, midtones, and highlights. However, to determine specific tones in shadows, the darkest areas of chosen reference should be examined. In this case, areas between the darkest and the brightest zones look slightly warm. That is why a red hue needs to be added to the shadows. Increased excessive red tones can be fixed by adding a bright greenish hue to midtones and blending the hues (50-60%) with each other.



Figure 41 Digital photo with the applied primary preset

After making the base preset further work will be done in the Dehancer plugin. Preset must be saved and the Dehancer opened. The new Dehancer preset is made from scratch.

The process should start with a grain generator. According to the reference, slide film has a small amount of fine film grain. The brightest areas look smooth and have no grain. On the contrary, Shadows have a slight amount of it. The grain in midtones looks balanced. The resolution of the image has to be moderately decreased to avoid the excessive sharpness of a digital image.

Kodak Endura Glossy Paper profile must be chosen in the “Print” section. Tonal Contrast should be added to emphasize the difference between the brightest and the darkest areas.

The next step is Halation. Slide film has a strong anti-halation layer. It means that halation is seen only in the brightest edges and has a small intensity in other areas of an image. Edges and spreading of halation can be controlled by using a Mask Mode.

Slide films do not have strong micro contrast in the brightest area. Bloom has to be added to slightly emphasize the brightest zones. Using a Mask Mode determines the brightest areas. Bloom has to be applied with low diffusion and low impact on the whole picture.

Further post-production process must be continued in Adobe Lightroom CC. To get rid of digital noise, noise reduction was applied. Final details have to be added in the Presence section to finish the picture. To restore micro contrast in shadows Clarity and Dehaze must be added. To add smoothness of 100 iso film Texture has to be decreased slightly.

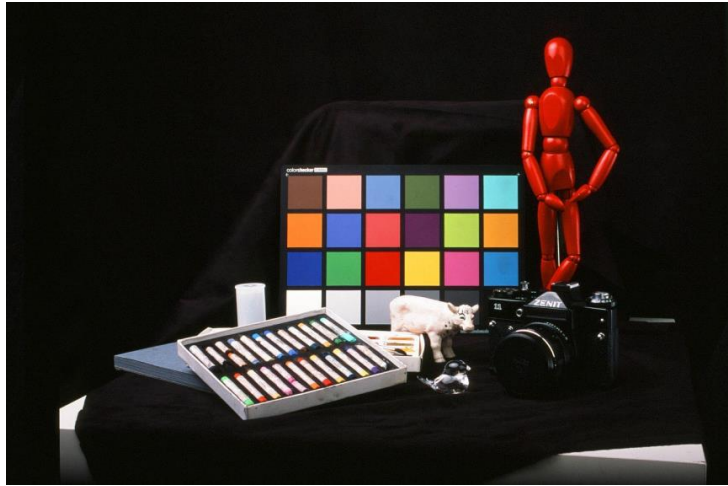


Figure 42 Fujichrome Provia 100F photo

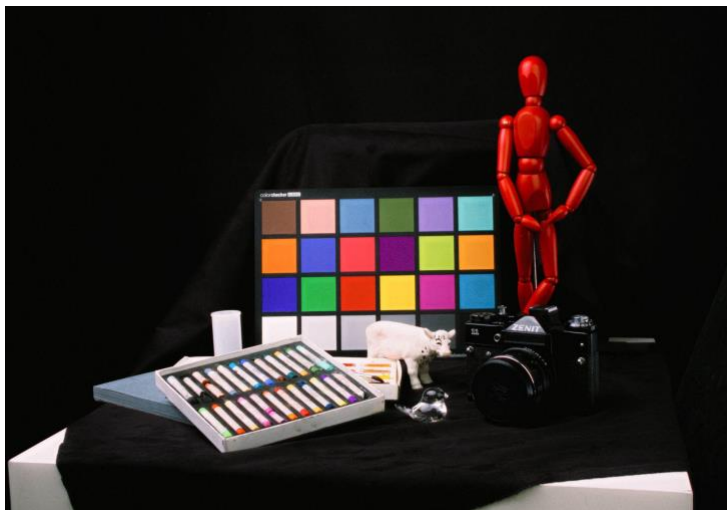


Figure 43 Digital photo after the post-production process

3.6 Reusability of presets on other photos

These Adobe Lightroom CC primary presets are not universal but can be reused in some cases. Applying the preset can change the colors of the picture to bizarre tones (Figure 45) but it can be fixed by changing the white balance and further settings (Figure 46). The presets were done to emulate daylight film in studio light conditions. If the color temperature differs from the studio lighting, the whole tonality will be ruined. For the Kodak Portra 400 preset, changing the white balance in most cases is enough to fix the problem. For the Fujifilm Provia 100f preset, shadows and highlights must be calibrated besides the white balance.



Figure 44 Default settings



Figure 45 The primary preset applied



Figure 46 The primary preset with fixed white balance

The Dehancer presets also can be reusable. However, two parameters should be controlled (Figure 47) most of the time when the preset is applied to an image. The first one is halation, the second one is Bloom. The impact of these parameters should be controlled according to the exposure of an image. If an image has contrast on the edges or bright artificial light, it is better to add more halation. Bloom could be added to the bright reflection of the light or during the imitation of intentional overexposing of the scene.

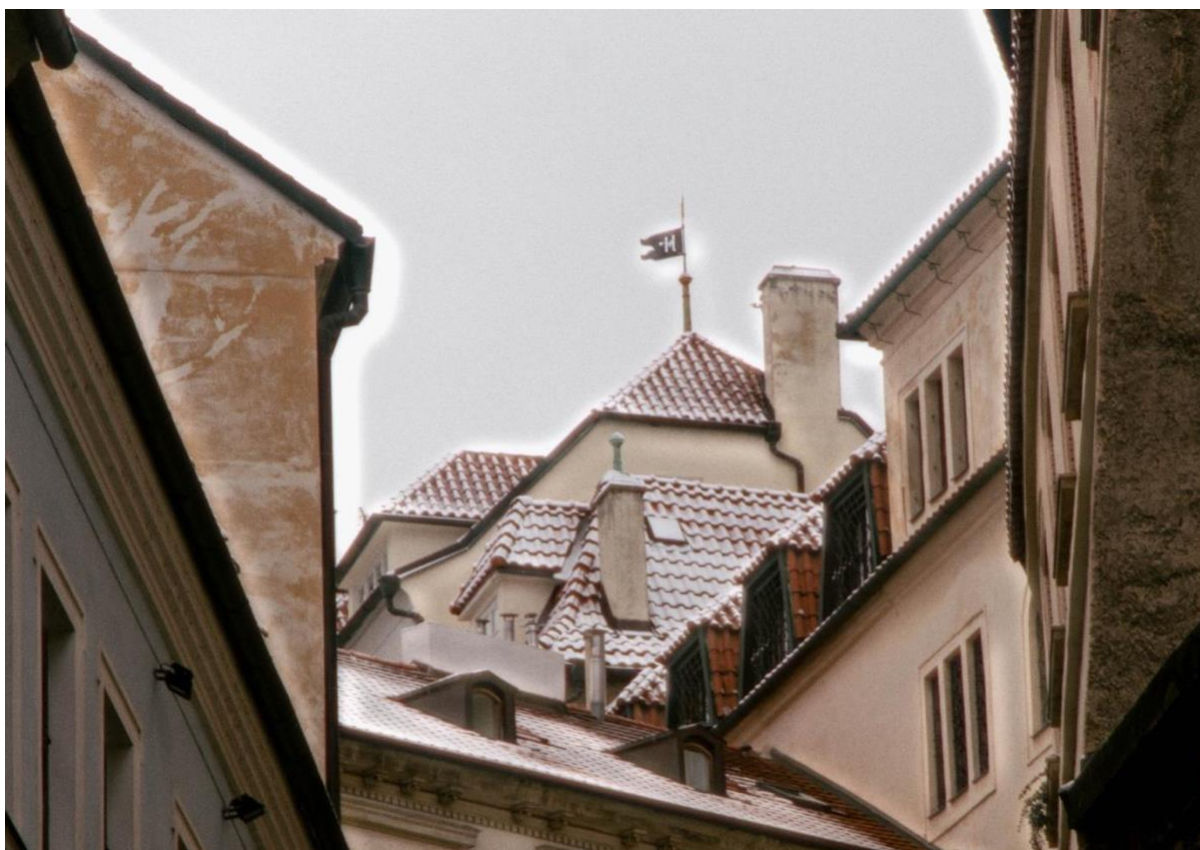


Figure 47 The example of uncontrolled halation and bloom effects

3.7 The results of the experiment

The results came out successful but there are details that do not respond to 100% analog image representation.

The main problems include working with color and shadows. This was expected due to the facts presented in the second chapter. For example, Color grading midtones affect the hues of dark objects (not in shadow). This can be seen in the figure ... Camera in the two pictures is different colors: one of them has a slight green tint, while another one has a red tint.

Sometimes software creates bugs. As seen in the figure ... bugs underline highlights in places where there should be none of them.

Using masks to correct these defects can help, however, this method can not be applied to the video footage.

In digital photos, shadows look unnatural. They do not blend into the lighter parts of the picture, there is no transition (figure ...).

On the other hand, some essential parts of the film look, like Bloom, came out quite impressive. Bloom creates a high contrast in the highlights of the photos.

With the development of the gear and software achieving this look becomes easier. The color grading section, which was necessary for this process, became available for Lightroom a few years ago. Reduction of the digital noise would not be possible without any image distortion.

Using a properly calibrated monitor is a crucial part of the successful results. Comparing two monitors (iMac 5k retina 2017 and MSI GE62 2QC Apache) showed that quality outcomes can be achieved only if a monitor is able to show the smallest details.

4. Conclusion

As the thesis shows, the popularity of film cameras and “film look” are on the rise. The existence of the Dehancer plugin confirms the high interest directed to the topic. The thesis is reacting to this circumstance by providing a workflow that could emulate different types of film stock.

In the first part, the thesis defines the key characteristics of film and digital images. The essential discoveries included the nature of digital noise and film grain, and the importance of small details like halation. The key detail that makes or breaks the film emulation is the transitions from the darkest to the brightest areas. It must be smooth. Moreover, the necessity for digital shots to be properly exposed can not be undermined, because this guarantees that most of the information will be preserved for the post-production. This knowledge is later used to develop a workflow for emulation.

The thesis argues that the main hypothesis (defined as “post-production process could emulate the look of film stocks upon the condition of using specific software and workflow”) was proven to be correct. During the post-production process, it was possible to recreate main characteristics of the film stocks: the Bloom from bright objects (red circle), the color rendering (blue circle), the halation (green circle), film grain (yellow circle) (Figure 48, Figure 49).

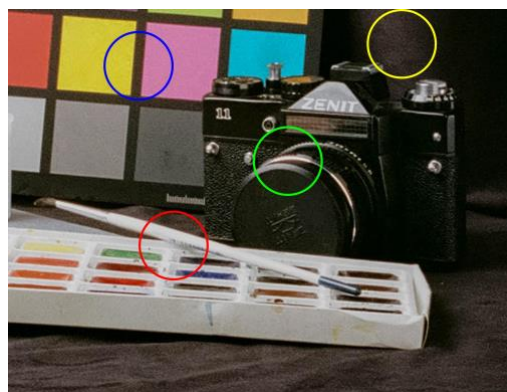


Figure 48 Kodak Portra 400 detail

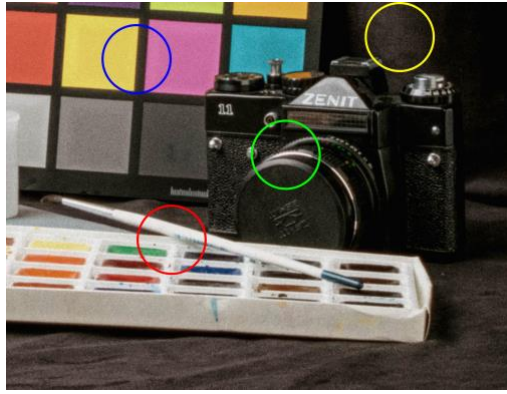


Figure 49 Digital photo after post-production detail

To sum up the findings of the practical part, the process of emulation should start with finding a reference picture. Exposure and final composition are one of the first things the author should keep in their mind before shooting.

Moreover, any author can not create a universal preset for all the pictures. Different lighting conditions and desired results call for an individual approach. Color representation can be influenced by the used lens. As the experiment showed, the vintage lens used created less contrast in the picture, which helped to emulate the color negative film stock look. On the other hand, this could be a disadvantage to emulate another type of film stock. This is another reason to consider all the variables beforehand. To conclude the practical findings, the thesis once again emphasizes the fact that a universal preset can not exist, however following the workflow can help achieve satisfactory results. No matter how good the results came out, the software still has its limitations. Any progress in the technology would make it possible to create more efficient workflows.

Future practitioners should consider using vintage lenses with a characteristic film look. For example, older Zeiss Planar lenses, Leica lenses, and Voigtlander lenses that were not available to the author during the time of the experiment. These lenses would help to achieve smooth boke on wide open aperture and soft tonality in color rendering, which are characteristic for analog old photos. One may also consider using gradient ND filters to create better results of exposure during shootings in daylight. As was mentioned before, correct exposure is a key factor to successful film emulation. The amount of data in a digital RAW file allows to get necessary information for further work in the post-production phase.

Returning to the contribution of the thesis, the workflow is made according to the authors observations of film properties and has its limitations. The gap in the solving of the problem could not be addressed properly since the matter of the experiment is creative expression. Any of the future practitioners can develop an equally usable workflow that suits their needs. However, the author of the thesis provides his experience to contribute to the broader spread of interest in the field of film emulating.

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